



```
In [1]: import spacy
import json
```

```
In [ ]: # file = "DATA/BASIL-main/articles/2019/0d10341a-9dba-4374-a524-814c300d1611_1
file = "DATA/BASIL-main/articles/2016/7a97de89-1433-46f7-bcc2-f80b041cb9a0_2.j
# file = "DATA/BASIL-main/articles/2016/d4a4bc34-e48b-4485-86b2-f1daf4d464dd_1
# file = "DATA/BASIL-main/articles/2015/2b9468af-4ab0-4f25-85f4-c3ef93d72006_1
def to_article_BASIL(filename: str) -> list[str]:
    ret: list[str] = []
    with open(filename, "r", encoding="utf-8") as f:
        data = json.load(f)

    #Export from nested list to a single list of sentences
    for paragraph in data["body-paragraphs"]:
        if len(paragraph) == 1:
            ret.append(paragraph[0])
        else:
            for sentence in paragraph:
                ret.append(sentence.lower())

    return ret
article = []
Sentence_list = to_article_BASIL(file)
#get only the first ten sentences
text = "".join(Sentence_list[:10]).lower()
text
```

```
Out[ ]: 'at almost all of his campaign rallies, donald j. trump leads a familiar refr
ain as he details his plan to build a giant wall along the border with mexic
o."who's gonna pay?" he asks.without fail, mr. trump's adoring fans reply, "m
exico!"while the mexican government has said little about mr. trump's plan to
beef up border security, two of the country's former presidents have a messag
e for mr. trump: mexico won't pay."i'm not going to pay," vicente fox said, u
sing a profanity to comment about the wall thursday in an interview with fusi
on."he should pay for it. he's got the money."mr. fox, who was president of m
exico from 2000 to 2006, was clear in his opposition to mr. trump, calling hi
m "egocentric" and lacking in any political values.mr. fox also expressed dis
belief that mr. trump did well with hispanic voters in the nevada caucuses on
tuesday and called on them to open their eyes."i'd like to know who those his
panics are," mr. fox said."they are followers of a false prophet."mr. fox's s
uccessor, felipe calderón, expressed similar concerns about the wall this mon
th."mexican people, we are not going to pay any single cent for such a stupid
wall!" mr. calderón told cnbc."and it's going to be completely useless."mr. c
alderón, who was president from 2006 until 2012, went on to argue that mr. tr
ump was misinformed about the flow of immigrants from mexico to the united st
ates and that closing the border would harm america's economy.for his part, m
r. trump has insisted that as president he would have the leverage to compel
mexico to pay for the wall, which he estimates would cost about $8 billion.'
```

```
In [37]: nlp = spacy.load("en_core_web_trf")
doc = nlp(text)
```

```
In [33]: methods = [ m for m in dir(doc) if not m.startswith('_')]
```

```
print(methods)
```

```
['cats', 'char_span', 'copy', 'count_by', 'doc', 'ents', 'extend_tensor', 'from_array', 'from_bytes', 'from_dict', 'from_disk', 'from_docs', 'from_json', 'get_extension', 'get_lca_matrix', 'has_annotation', 'has_extension', 'has_unknown_spaces', 'has_vector', 'is_nerred', 'is_parsed', 'is_sentenced', 'is_tagged', 'lang', 'lang_', 'mem', 'noun_chunks', 'noun_chunks_iterator', 'remove_extension', 'retokenize', 'sentiment', 'sents', 'set_ents', 'set_extension', 'similarity', 'spans', 'tensor', 'text', 'text_with_ws', 'to_array', 'to_bytes', 'to_dict', 'to_disk', 'to_json', 'to_utf8_array', 'user_data', 'user_hooks', 'user_span_hooks', 'user_token_hooks', 'vector', 'vector_norm', 'vocab']
```

```
In [38]: for ent, chuck in zip(doc.ents, doc.noun_chunks):  
         print(ent.text, ent.label_, chuck.text)
```

```
donald j. trump PERSON almost all  
mexico GPE his campaign rallies  
trump PERSON donald j. trump  
mexican NORP a familiar refrain  
trump PERSON he  
two CARDINAL his plan  
trump PERSON a giant wall  
mexico GPE the border  
vicente fox PERSON mexico  
thursday DATE he  
fusion ORG fail  
fox PERSON mr. trump's adoring fans  
mexico GPE the mexican government  
2000 DATE mr. trump's plan  
2006 DATE border security  
trump PERSON the country's former presidents  
fox PERSON a message  
trump PERSON mr. trump  
hispanic NORP mexico  
nevada ORG vicente fox  
tuesday DATE a profanity  
hispanics NORP the wall  
fox PERSON an interview  
fox PERSON fusion  
felipe calderón PERSON it  
this month DATE he  
any single cent MONEY the money  
calderón PERSON . fox  
cnbc ORG who  
calderón PERSON president  
2006 DATE mexico  
2012 DATE his opposition  
trump PERSON mr. trump  
mexico GPE him  
the united states GPE . fox  
america GPE disbelief  
trump PERSON mr. trump  
mexico GPE hispanic voters  
about $8 billion MONEY the nevada caucuses
```

```
In [39]: # for token in doc:
#         print(token.text, token.lemma_, token.pos_, token.dep_)

# for ent, chuck in zip(doc.ents, doc.noun_chunks):
#         print(ent.text, ent.label_, chuck.text)

Name_TAG = {(ent.text, ent.label_) for ent in doc.ents if ent.label_ == "PERSON"}
print(Name_TAG)
```

```
{('the united states', 'GPE'), ('fox', 'PERSON'), ('trump', 'PERSON'), ('hispanic', 'NORP'), ('vicente fox', 'PERSON'), ('mexican', 'NORP'), ('calderón', 'PERSON'), ('donald j. trump', 'PERSON'), ('america', 'GPE'), ('mexico', 'GPE'), ('hispanics', 'NORP'), ('felipe calderón', 'PERSON')}
```

```
In [40]: for chunk in doc.noun_chunks:
          print(chunk.text)
```

almost all
his campaign rallies
donald j. trump
a familiar refrain
he
his plan
a giant wall
the border
mexico
he
fail
mr. trump's adoring fans
the mexican government
mr. trump's plan
border security
the country's former presidents
a message
mr. trump
mexico
vicente fox
a profanity
the wall
an interview
fusion
it
he
the money
. fox
who
president
mexico
his opposition
mr. trump
him
. fox
disbelief
mr. trump
hispanic voters
the nevada caucuses
tuesday
them
their eyes
who
those hispanics
mr. fox
followers
a false prophet
. fox's successor
felipe calderón
similar concerns
the wall
we
any single cent
such a stupid wall

mr. calderón
cnbc
it
. calderón
who
president
mr. trump
the flow
immigrants
mexico
the united states
the border
america
his part
mr. trump
president
he
the leverage
mexico
the wall
which
he

In []:

president compound trump
trump nsubj stood
stood ROOT stood
firm oprd stood
friday npadvmod stood
on prep stood
his poss demands
demands pobj on
for prep demands
a det wall
border compound wall
wall pobj for
after mark broke
the det meeting
second amod meeting
white compound house
house compound meeting
meeting nsubj broke
with prep meeting
congressional amod leaders
leaders pobj with
this det week
week npadvmod meeting
broke advcl stood
up prt broke
with prep broke
no det deal
apparent amod deal
deal pobj with
, punct stood
warning advcl stood
democrats dobj warning
the det shutdown
partial amod shutdown
government compound shutdown
shutdown nsubj last
could aux last
last ccomp warning
" punct last
years npadvmod last
" punct last
and cc warning
saying conj warning
he nsubj declare
could aux declare
even advmod declare
declare ccomp saying
a det emergency
" punct emergency
national amod emergency
emergency dobj declare
" punct emergency
to aux bypass
bypass advcl declare

congress dobj bypass
if mark necessary
necessary advcl bypass
. punct stood
"we nsubj call
can aux call
call ccomp told
a det emergency
national amod emergency
emergency dobj call
[punct emergency
to aux build
build acl emergency
a det wall
border compound wall
wall dobj build
] punct call
because prep call
of pcomp because
the det security
security pobj because
of prep security
our poss country
country pobj of
, punct told
" punct told
trump nsubj told
told ROOT told
reporters dobj told
in prep told
the det garden
rose compound garden
garden pobj in
, punct told
during prep told
a det conference
lengthy amod conference
and cc lengthy
impromptu conj lengthy
press compound conference
conference pobj during
. punct told
"i nsubj do
may aux do
do ccomp said
it dobj do
, punct said
" punct said
he nsubj said
said ROOT said
, punct said
before prep said
adding pcomp before
, punct adding

" punct adding
if mark do
we nsubj do
can aux do
do advcl giving
it dobj do
through prep do
a det process
negotiated amod process
process pobj through
, punct giving
we nsubj giving
're aux giving
giving ccomp adding
it dative giving
a det shot
shot dobj giving
. punct said
"the det conference
press compound conference
conference ROOT conference
underscored ROOT underscored
how advmod far
far advmod apart
apart acomp are
both det sides
sides nsubj are
are ccomp underscored
, punct underscored
even advmod called
as mark called
trump nsubj called
called advcl underscored
the det meeting
meeting dobj called
" punct productive
productive oprd called
" punct productive
and cc called
suggested conj called
the det standoff
standoff nsubj end
could aux end
end ccomp suggested
soon advmod end
-- punct end
or cc end
not conj end
. punct underscored
he nsubj indicated
indicated ROOT indicated
he nsubj shifting
was aux shifting
not neg shifting

shifting ccomp indicated
on prep shifting
his poss demand
demand pobj on
for prep demand
more amod billion
than quantmod billion
\$ quantmod billion
5 compound billion
billion pobj for
for prep billion
funding pobj for
for prep funding
a det wall
wall pobj for
on prep wall
the det border
southern amod border
border pobj on
, punct indicated
saying advcl indicated
it nsubj was
was ccomp saying
necessary acomp was
as mark is
the det border
border nsubj is
is advcl was
a det disaster
" punct disaster
dangerous amod disaster
, punct disaster
horrible amod disaster
disaster attr is
. punct indicated
""this punct told
is ccomp talking
national amod security
security attr is
we nsubj talking
're aux talking
talking relcl security
about prep talking
, punct talking
we nsubj talking
're aux talking
not neg talking
talking ccomp told
about prep talking
games pobj about
, punct told
" punct told
he nsubj told
told ROOT told

reporters dobj told
. punct told
" punct have
we nsubj playing
're aux playing
not neg playing
playing ccomp have
games dobj playing
, punct have
we nsubj have
have ROOT have
to aux do
do xcomp have
it dobj do
. punct have
"despite ROOT "despite
trump nsubj saying
saying pcomp "despite
conversations nsubj come
had aux come
" punct come
come ccomp saying
a det way
long amod way
way dobj come
, punct were
" punct were
democrats nsubj were
were ROOT were
more advmod dour
dour acomp were
in prep dour
their poss assessment
assessment pobj in
. punct were
senate compound leader
minority compound leader
leader compound schumer
chuck compound schumer
schumer nsubj exited
exited ROOT exited
saying advcl exited
trump nsubj told
told ccomp saying
them dobj told
" punct told
he nsubj keep
'd aux keep
keep ccomp told
the det government
government nsubj closed
closed ccomp keep
for prep keep
a det period

very advmod long
long amod period
period pobj for
of prep period
time pobj of
, punct period
months appos period
or cc months
even advmod years
years conj months
. punct exited
"trump punct exited
later advmod confirmed
confirmed ROOT confirmed
in prep confirmed
the det garden
rose compound garden
garden pobj in
, punct confirmed
" punct confirmed
absolutely advmod said
i nsubj said
said ccomp confirmed
that dobj said
, punct confirmed
" punct confirmed
while mark clarifying
clarifying advcl confirmed
he nsubj hopes
hopes ccomp clarifying
the det shutdown
partial amod shutdown
shutdown nsubj last
does aux last
n't neg last
last ccomp hopes
more amod few
than quantmod few
a quantmod few
few nummod days
more amod few
days npadvmod last
. punct confirmed
he nsubj said
said ROOT said
it nsubjpass opened
could aux opened
be auxpass opened
opened ccomp said
" punct opened
very advmod quickly
quickly advmod opened
" punct opened
if mark come

they nsubj come
come advcl opened
to prep come
an det agreement
agreement pobj to
on prep agreement
the det wall
wall pobj on
. punct said
he nsubj offered
offered ROOT offered
some det flexibility
flexibility dobj offered
only advmod on
on prep flexibility
what pobj of
the det wall
wall nsubjpass made
would aux made
be auxpass made
made pcomp on
of prep made
, punct offered
saying advcl offered
it nsubj be
could aux be
be ccomp saying
concrete attr be
or cc concrete
steel conj concrete
. punct offered
"steel nsubj is
is ccomp said.yet
more advmod expensive
expensive acomp is
than prep expensive
the det concrete
concrete pobj than
, punct is
but cc is
i nsubj feel
feel conj is
like mark talking
we nsubj talking
're aux talking
talking advcl feel
about prep talking
steel pobj about
because mark feels
the det side
other amod side
side nsubj feels
feels advcl feel
better acomp feels

about prep better
it pobj about
, punct is
" punct is
he nsubj said.yet
said.yet ROOT said.yet
house compound speaker
speaker compound pelosi
nancy compound pelosi
pelosi nsubj said
said ROOT said
the det democrats
democrats poss view
' case democrats
view nsubj is
is ccomp said
that mark resolve
they nsubj resolve
can aux resolve
not neg resolve
resolve ccomp is
the det dispute
funding compound dispute
dispute dobj resolve
until mark opens
the det government
government nsubj opens
fully advmod opens
opens advcl resolve
. punct said
she nsubj said
said ROOT said
there expl been
had aux been
been ccomp said
progress attr been
only advmod in
in prep progress
the det sense
sense pobj in
of prep sense
a det understanding
" punct understanding
better amod understanding
understanding pobj of
" punct understanding
of prep understanding
each det other
other pobj of
. punct said